AHCI PRACTICE QUESTIONS WITH ANSWERS

Q1) Suppose a UI designer designs project of abc company, he performs all the best practices, following all the regulations strictly from start to end without including other colleagues, at run time he successfully presents it to the end terminal, is this cognition process relates to semantic memory?

A)Semantic memory is derived from the episodic memory, the cognition process involves both semantic and episodic. (reference: chp1 slides 19 to 20).

Q2) A student is required to write a biography of a famous personality, the student studied the personality earlier but due to number of assignments and tasks he forgets a part of theory, after some practice he manages to extract that lost part of memory back and becomes able to apply it in a required manner. What are that practices that lead to get back the lost memory part again?

A) Recall and recognition (reference: slides 27 and 28, chp 1).

Q3) A software engineer studies the requirements carefully and design a fully functional vehicle monitoring application with serious intention but at the run time the system shows wrong location of the car, the PM indicates this happening as a human error, mention the type of error and explain why?

A)Slips (chp1 slides, 37, 38).

CHAPTER 3 (TAKE OUTS)

WIMP

WIMP supports **direct manipulation**, visual feedback, and ease of learning, making it widely used in desktop operating systems.

Explain two usability advantages and two limitations of WIMP interfaces.

Advantages

1. **Learnability** — visual representation supports recognition over recall.

2. **Immediate feedback** — user actions produce visible results.

Limitations

1. **Screen space dependency** — many windows/icons can cause clutter.

2. **Not suitable for immersive or spatial tasks** — limited support for depth and real-world interaction.

Compare WIMP interfaces and 3D interfaces based on interaction style and user cognition.

Aspect	WIMP Interface	3D Interface
Interaction	Mouse and pointer based	Spatial navigation and gestures
Representation	2D graphical objects	Virtual 3D environment
Cognitive Load	Lower, familiar metaphors	Higher, requires spatial understanding
Application	Office systems, desktops	VR, simulations, gaming

3D interfaces provide immersion but demand more perceptual and cognitive processing.

Describe two challenges users face when interacting with 3D interfaces.

1. **Spatial disorientation** — users may lose orientation in virtual space.

2. **Navigation complexity** — controlling movement in three dimensions requires additional cognitive effort and motor coordination.

These challenges increase cognitive load compared to traditional WIMP interfaces.

Why are WIMP interfaces considered examples of direct manipulation? Explain.

WIMP interfaces support **direct manipulation** because users interact with visible objects using physical actions such as pointing, dragging, and selecting. The system provides:

- continuous visual feedback,
- reversible actions,
- and intuitive object representation.

This reduces memory load and improves usability.

Provide one real-world application where a 3D interface is more suitable than a WIMP interface and justify.

Medical surgery simulation systems use 3D interfaces because tasks involve spatial depth, hand movement coordination, and realistic visualization. A WIMP interface cannot effectively represent real-world spatial relationships required for training.

WIMP = efficiency and familiarity; 3D = immersion but higher cognitive demand.

Spatial Disorientation — Real World Example

Example: First-time user in a VR building walkthrough

A student wears a headset and explores a virtual university building. After moving through several corridors and rotating views, they lose sense of direction and cannot tell:

- where the entrance was
- which way is forward
- how to return to the starting point

This happens because the brain receives **visual motion without physical movement**, disrupting spatial orientation cues (balance + body feedback).
The user feels “lost” even though the interface is working correctly.

Navigation Complexity — Real World Example

Example: 3D medical imaging software

A trainee doctor uses a 3D anatomy viewer to inspect an organ. To reach the correct view, they must:

- rotate the model
- zoom
- pan
- adjust viewing angle

Coordinating these controls simultaneously is cognitively demanding. The user understands the goal but struggles with **how to move efficiently in 3D space**.

One-Line Contrast

Spatial disorientation = “I don’t know where I am.”
Navigation complexity = “I know where I want to go, but movement is hard.”

CHAPTER 4 TAKE OUTS

Q1) The development of Sketchpad by Ivan Sutherland is considered a milestone in the history of Human–Computer Interaction.

Explain how the work of a single creative researcher contributed to the evolution of interactive computing. Support your answer with examples from modern computer systems.

A1) The work of Ivan Sutherland demonstrates how a single creative individual can significantly influence the development of computing technology. In 1963, Sutherland developed Sketchpad, one of the first interactive computer graphics systems.

Before Sketchpad, computers were mainly used through text-based commands and punched cards. Users could not directly interact with graphical objects on a screen. Sketchpad introduced the concept of interactive graphics, allowing users to draw shapes on the screen using a light pen and manipulate objects directly.

Q2) Explain how Alan Kay was influenced by the ideas of Douglas Engelbart and Seymour Papert in designing an interactive computing system.

In your answer, categorize the contributions of Engelbart and Papert and explain how these ideas shaped Alan Kay's system design in the context of Human–Computer Interaction.

A2) 1. Influence of Douglas Engelbart (Interactive Computing)

Douglas Engelbart focused on augmenting human intellect through computers. He developed interactive technologies that allowed users to directly interact with digital information.

Key contributions include:

- the computer mouse
- graphical interaction with on-screen objects
- collaborative computing concepts

Engelbart's work demonstrated that computers should be used as interactive tools rather than passive calculation machines. This idea influenced Alan Kay to design systems where users could directly interact with digital objects.

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2. Influence of Seymour Papert (Learning and Education)

Seymour Papert emphasized learning through exploration and creativity using computers. His work on the Logo programming language encouraged children to learn by experimenting with programming and graphics.

Papert believed computers should support active learning and creativity rather than simply delivering information.

This philosophy inspired Alan Kay to think about computers as personal learning tools, especially for children.

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3. Alan Kay's System Design

Combining these influences, Alan Kay developed the concept of the Dynabook.

The Dynabook concept included:

- a portable personal computer
- graphical user interface
- interactive multimedia learning
- object-oriented programming environment

Q3) WYSIWYG

A3) WYSIWYG means the computer screen shows the final result while you are editing, so what you see is exactly what you get in the output.

Q4) WYSIWYG is not a panacea. What is panacea?

A4) A panacea is a remedy or solution that is believed to cure all problems or illnesses, though in reality, such universal cures rarely exist.

CHAPTER 5 TAKE OUTS

Ethnography is a branch of anthropology and the systematic study of individual cultures. It explores cultural phenomena from the point of view of the subject of the study. Ethnography is also a type of social research that involves examining the behavior of the participants in a given social situation and understanding the group members' own interpretation of such behavior.

Ethnographers mainly use qualitative methods, though they may also include quantitative data. The typical ethnography is a holistic study and so includes a brief history, and an analysis of the terrain, the climate, and the habitat. A wide range of groups and organisations have been studied by this method, including traditional communities, youth gangs, religious cults, and organisations of various kinds. While, traditionally, ethnography has relied on the physical presence of the researcher in a setting, there is research using the label that has relied on interviews or documents, sometimes to investigate events in the past such as the NASA *Challenger* disaster. There is also ethnography done in "virtual" or online environments, sometimes labelled netnography or cyber-ethnography.

A **lintel** or **lintol** is a type of beam (a horizontal structural element) that spans openings such as portals, doors, windows and fireplaces. It can be a decorative architectural element, or a combined ornamented/structural item. In the case of windows, the bottom span is referred to as a sill, but, unlike a lintel, does not serve to bear a load to ensure the integrity of the wall.

Persona Example

Name

Ali Khan

Age

21 years old

Occupation

3rd-year BS Computer Science student

Technical Skill

Intermediate computer user

Goals

- Quickly download lecture slides
- Submit assignments before deadlines
- Check grades and announcements easily

Frustrations

- Too many confusing menu options
- Difficult to find assignment submission page
- Slow navigation between course sections

Typical Usage Scenario

Ali usually accesses the portal late at night from his laptop or phone to:

1. Download lecture slides
2. Upload assignments
3. Check deadlines for different courses

If the interface is complex or information is hidden inside multiple menus, Ali becomes frustrated and wastes time searching.

How Persona Helps in HCI Design

Using this persona, designers can make better decisions:

- **Simplify navigation** so assignments are easy to locate
- Provide **clear deadlines and notifications**
- Ensure the system works well on **mobile devices**

By designing for Ali's needs, the system becomes easier for most students to use.

PROCESS OF DESIGN EXAMPLE:

Real-World Example: Designing a University Library Mobile App

1. Identify Needs and Establish Requirements

The university wants a **mobile app for the library**.

Designers first gather requirements by interviewing students and librarians.

Findings

- Students want to **search books quickly**.
- They want to **check book availability**.
- They want **notifications for return deadlines**.

This stage focuses on understanding **user needs and context of use**.

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2. Develop Alternative Designs

Designers propose multiple interface ideas.

Example alternatives:

- Design A: Search bar on the homepage.
- Design B: Category-based browsing (subjects like Computer Science, Physics).
- Design C: Voice search for books.

Different ideas are explored before choosing the best solution.

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3. Build Interactive Prototypes

Designers create a **prototype** of the app.

Examples:

- paper sketches of screens
- clickable digital mockups

Students test the prototype to see how the interface works.

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4. Evaluate the Design

Students are asked to perform tasks such as:

- searching for a book
- reserving a book
- checking due dates

Designers observe problems such as:

- students cannot easily find the reservation button
- navigation is confusing

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5. Iterate and Improve

Based on feedback, the design is improved:

- the reservation button is made larger
- navigation is simplified

- notifications are added

The cycle continues until usability improves.

CHAPTER 6 TAKE OUTS

USABILITY ATTRIBUTES

A software development team is designing a **UAV monitoring dashboard** for engineers. They want to ensure that the system is easy to use and efficient. To evaluate usability, they focus on the task:
"An operator must detect and respond to a wind disturbance alert within the system."

They define the following:

- **Measuring Concept:** Time taken to respond to an alert
- **Measuring Method:** Observing users in a test environment and recording response times
- **Now Level:** 12 seconds (current prototype)
- **Worst Case:** 20 seconds
- **Best Case:** 5 seconds
- **Planned Level:** 8 seconds

Question:
Explain each of the above usability attributes in the context of the scenario. Why is it important to define all these levels when evaluating usability?

ANSWER:

In this scenario, the team is evaluating usability using structured attributes:

Explanation of Usability Attributes

1. **Measuring Concept**
   This defines *what* aspect of usability is being evaluated.
   - Here: **Time taken to respond to an alert** (efficiency).

2. **Measuring Method**
   This defines *how* the concept is measured.
   - Here: Observing users and recording response time in a controlled test.
3. **Now Level (Current Level)**
   The current performance of the system.
   - Here: Users take **12 seconds** on average.
4. **Worst Case**
   The lowest acceptable performance limit.
   - Here: **20 seconds** is considered too slow and unacceptable.
5. **Best Case**
   The ideal or optimal performance.
   - Here: **5 seconds** represents excellent usability.
6. **Planned Level (Target Level)**
   The desired usability goal set by designers.
   - Here: The team aims for **8 seconds**.

Importance of Defining These Levels

- **Provides Clear Evaluation Criteria**
  Designers know whether the system is acceptable or needs improvement.
- **Guides Design Decisions**
  Helps prioritize improvements to reach the planned level.
- **Supports Objective Measurement**
  Removes guesswork by using quantifiable data.
- **Tracks Progress Over Time**
  Teams can compare the *now level* with the *planned level*.
- **Ensures User-Centered Design**
  Focuses on real user performance rather than assumptions.

Problem (Usability Measurement)

A team is evaluating the usability of a mobile application for emergency response. They measure the **time (in seconds)** taken by 5 users to complete a critical task:

- User 1: 10 sec
- User 2: 12 sec
- User 3: 9 sec

- User 4: 11 sec
- User 5: 8 sec

The usability targets are defined as:

- **Worst Case:** 15 sec
- **Best Case:** 7 sec
- **Planned Level:** 9 sec

Question:

1. Calculate the **average (mean) task completion time**.
2. Compare the result with the **planned level**.
3. Determine whether the system meets the usability goal or needs improvement.

Solution

Step 1: Calculate the Average Time

Average=(10+12+9+11+8)/5

Average=50/5=10 seconds

Step 2: Compare with Planned Level

- **Average Time = 10 sec**
- **Planned Level = 9 sec**

The current performance is **1 second slower** than the target.

Step 3: Evaluation

- The system is **better than the worst case (15 sec)**
- The system is **not meeting the planned level (9 sec)**
- It is also **far from the best case (7 sec)**

Final Conclusion

The system is **acceptable but not optimal**.
It **needs improvement to reach the planned usability level.**

Insight

Even small differences (like 1 second) can be important in usability, especially in time-critical systems such as emergency response or UAV monitoring.

TAKE OUTS (CHAPTER 7)

Question 1: Principles of Usability (Learnability, Flexibility, Robustness)

Scenario-Based Question

A software company develops an online food delivery mobile app. New users can easily place orders within minutes. The app allows multiple ways to search for food (by name, category, or voice search). It also provides clear error messages and allows users to track their orders in real time.

Question:
Analyze the application based on the three usability principles: learnability, flexibility, and robustness. Explain how each principle is reflected in the scenario.

Answer

The application demonstrates the three usability principles as follows:

Learnability refers to how easy it is for new users to begin interacting with the system.

- **In the scenario, users can place orders within minutes, indicating that the interface is simple and intuitive.**
- **Minimal training is required → high learnability.**

2. Flexibility

Flexibility refers to the variety of ways a user and system can exchange information.

- **The app allows users to:**
  - **Search by food name**
  - **Browse by category**
  - **Use voice search**
- **Multiple interaction methods → high flexibility.**

3. Robustness

Robustness refers to the level of support provided to the user to achieve goals.

- **The app provides:**
  - **Clear error messages**
  - **Order tracking system**
- **This helps users understand system status and recover from errors → strong robustness.**

Conclusion

The app is well-designed because it:

- **Is easy to learn (learnability)**
- **Offers multiple interaction options (flexibility)**
- **Provides strong user support and feedback (robustness)**

Question 2: Principles of Learnability

Scenario-Based Question

A university introduces a new Learning Management System (LMS) for students. The system uses familiar icons (like a folder for files), allows students to predict outcomes of actions (e.g., clicking "Submit" uploads assignments), provides feedback after every action, maintains consistent layout across pages, and allows students to apply knowledge learned in one section to another.

Question:
Evaluate the system based on the learnability principles: predictability, synthesizability, familiarity, generalizability, and consistency.

Answer

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1. Predictability

Users can anticipate the result of their actions.

- **Clicking "Submit" uploads assignments → expected behavior.**
- **System actions are predictable.**

2. Synthesizability

Users can understand the effect of past actions.

- **After submitting, the system shows confirmation (e.g., "Assignment uploaded").**
- **Users can assess what has happened.**

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3. Familiarity

The system uses concepts users already know.

- **Folder icons for files resemble real-world objects.**

- **Reduces learning effort.**

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4. Generalizability

Knowledge from one part of the system can be applied to others.

- **If students learn how to upload assignments in one course, they can do it in all courses.**
- **Promotes efficient learning.**

5. Consistency

Similar actions behave the same across the system.

- **Layout, buttons, and navigation remain uniform across pages.**
- **Reduces confusion.**

Conclusion

The LMS is highly learnable because it:

- **Matches user expectations (predictability)**
- **Provides feedback (synthesizability)**
- **Uses familiar concepts (familiarity)**
- **Supports knowledge transfer (generalizability)**
- **Maintains uniform design (consistency)**

Case Study Question

Learnability Principles: Predictability, Synthesizability, Familiarity, Generalizability, Consistency

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Case Study

A university has implemented a new **Learning Management System (LMS)** for students and faculty.

- The system uses **common icons** such as folders for files and a paper-plane icon for submission.
- When a student clicks the **"Submit Assignment"** button, the system behaves as expected and uploads the file.
- After submission, the system displays a **confirmation message** and allows students to view submitted work.
- The interface design, navigation menus, and button placements remain **consistent across all courses**.
- Skills learned in one course (e.g., uploading assignments) can easily be applied in other courses.

Question

Critically evaluate the LMS using the five principles of learnability:

1. **Predictability**
2. **Synthesizability**
3. **Familiarity**
4. **Generalizability**
5. **Consistency**

In your answer:

- Define each principle
- Apply each principle to the case study with examples
- Identify any weaknesses (if any)
- Suggest improvements to enhance learnability

Answer

Predictability

- Definition: Ability to predict system response
- Application: “Submit” behaves as expected
- Improvement: Add preview before submission

Synthesizability

- Definition: Ability to understand effects of actions
- Application: Confirmation message after submission
- Improvement: Add activity history log

Familiarity

- Definition: Use of real-world concepts
- Application: Folder and submission icons
- Improvement: Add tooltips for new users

Generalizability

- Definition: Transfer of knowledge across system
- Application: Same upload process in all courses
- Improvement: Provide shortcuts for frequent tasks

Consistency

- Definition: Uniform design and behavior
- Application: Same layout across LMS
- Improvement: Ensure consistency on mobile version

Conclusion

- The LMS strongly supports learnability but can be further improved with additional guidance and features.